JOSIAH SAJI

Robotics & ROS 2 Engineer

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PROFESSIONAL SUMMARY

Robotics Engineer specializing in ROS 2 (Jazzy/Humble), autonomous mobile robots (SLAM, Nav2), embedded firmware (ESP32, FreeRTOS, micro-ROS), and multi-DOF robot kinematics. Proven track record transitioning autonomous systems from mathematical formulation and dynamic simulation (Gazebo, RViz2) to real hardware prototypes.

WORK EXPERIENCE

Robotics Engineer — Jan 2026 – Present

Evolve Robotics · Kerala, India

- Contributing to ROS 2 mobile robot software integration, URDF/Xacro kinematic modeling, and system bringup launch systems.
- Configuring autonomous navigation pipelines, sensor parameters, and Gazebo simulation environments.
- Developing microcontroller firmware and hardware-in-the-loop test routines for mobile robotics platforms.

Secretary — Robotics Society — 2024 – 2025

Saintgits College of Engineering · Kottayam, Kerala

- Organized and facilitated hands-on technical workshops in ROS 2, micro-ROS, and microcontroller programming for undergraduate engineering students.
- Coordinated hardware procurement, prototyping components, and student technical mentorship.

KEY ENGINEERING PROJECTS

ROS 2 Quadruped Locomotion Framework — ROS 2 Humble · C++ / Python · Gazebo

- Formulated closed-form analytical 3-DOF leg inverse kinematics (~25–26 µs CPython 3.12 microbenchmark per leg solve) providing direct trigonometric solutions without iterative Jacobian inversion.
- Implemented parametric 3D cycloid foot trajectory generator guaranteeing **C^1 velocity continuity** and zero horizontal/vertical velocity boundary conditions at touchdown.
- Validated multi-gait sequencing (Trot, Walk, Crawl, Bound) and IMU active roll/pitch balance loop at 100 Hz in Gazebo.

Autonomous Tilling Robot (AgriTech) — ROS 2 Humble · Raspberry Pi 4B · ESP32 · LiFePO4

- Architected hybrid 24V agricultural rover with front 2-stroke engine and rear high-torque drive; developed and submitted for **YIP 7.0**.
- Integrated GPS waypoint navigation (NEO-6M), multi-sensor power telemetry, and wireless emergency RF teleoperation.
- Engineered custom chassis and LiFePO4 battery distribution architecture for high-vibration agricultural environments.

ROS 2 Jazzy 2WD Differential Drive Robot — ROS 2 Jazzy · Ubuntu 24.04 · Gazebo Harmonic · Nav2

- Developed modular navigation framework for 2WD + dual-castor chassis with RPLidar A1 M8 (360° FOV, 8m configured mapping range, 5.5–10 Hz).
- Configured SLAM Toolbox 2D occupancy mapping, Nav2 dynamic costmaps, and global/local path planning in Gazebo Harmonic.
- Authored multi-module curriculum covering URDF/Xacro, TF2 trees, RViz2, and automated bash installation.

ESP32 micro-ROS 2WD Differential Drive — ESP32 · FreeRTOS · micro-ROS (UDP) · C++

- Built native micro-ROS node on ESP32 executing command subscriptions over Wi-Fi UDP with micro-ROS agent.
- Implemented 4-pin pure PWM motor driver with static friction deadband cutoff (MIN_PWM = 60) and 1 kHz LEDC switching.
- Integrated 500 ms watchdog failsafe shutting off motors on Wi-Fi packet drop or agent disconnection; benchtop tested.

TECHNICAL SKILLS

Robotics & Autonomy:	ROS 2 (Jazzy, Humble), micro-ROS, Nav2, SLAM Toolbox, MoveIt 2, ros2_control, TF2
Simulation & CAD:	Gazebo (Harmonic, Classic 11), RViz2, URDF, Xacro, 3D Prototyping (STL/CAD)
Embedded & Hardware:	ESP32, Raspberry Pi, FreeRTOS, RPLidar A1, MPU6050/BNO055 IMU, LiFePO4 BMS, PWM/UART/I2C/SPI
Languages & Tools:	C++, Python, Embedded C, Linux (Ubuntu 24.04/22.04 LTS), Git, Bash, CMake

EDUCATION

Bachelor of Technology (B.Tech) in Robotics & Automation — 2021 – 2025

Saintgits College of Engineering · APJ Abdul Kalam Technological University, Kerala